

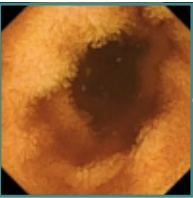
Lane A: passive-video forecasting

validated across three domains



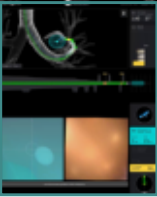
CholecT50 (laparoscopy)

last observed frame



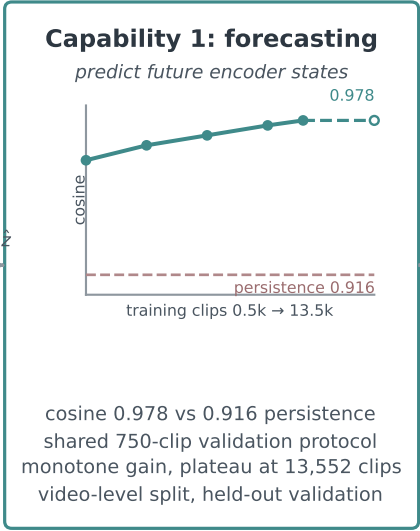
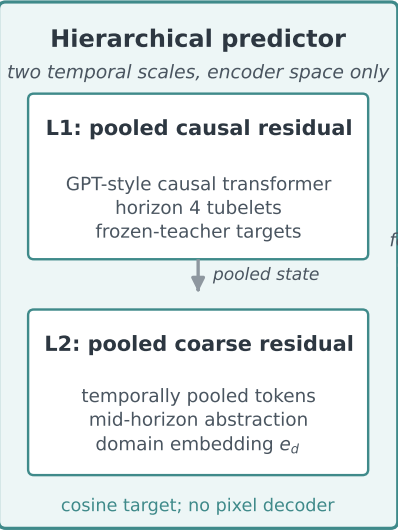
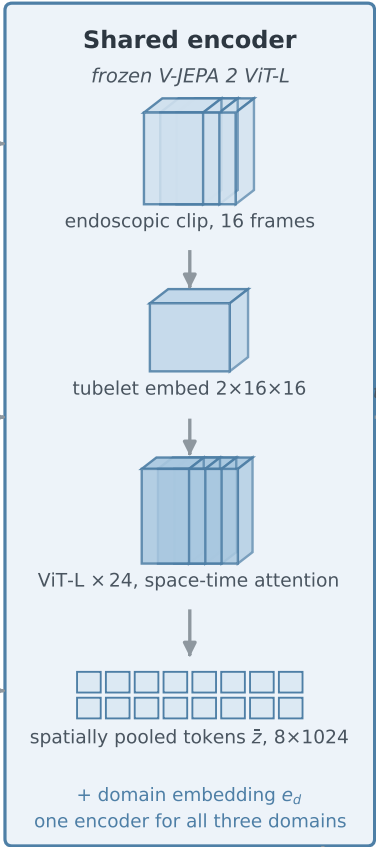
Kvasir-Capsule (GI)

last observed frame



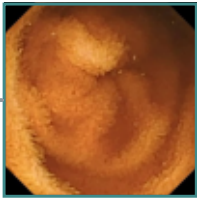
ION (bronchoscopy)

navigation-console capture



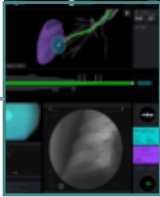
CholecT50 (laparoscopy)

predicted-latent retrieval



Kvasir-Capsule (GI)

predicted-latent retrieval



ION (bronchoscopy)

predicted-latent retrieval

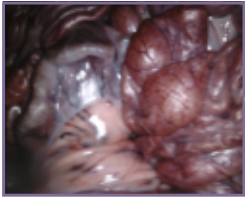
loss-of-view warning

camera-handling training

cross-domain representation

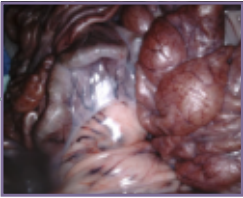
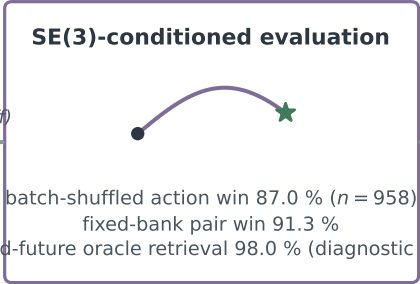
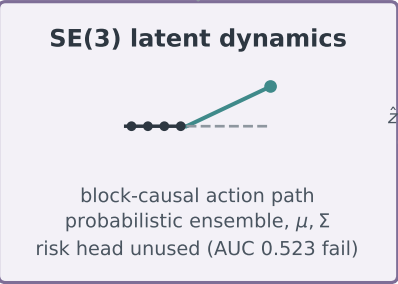
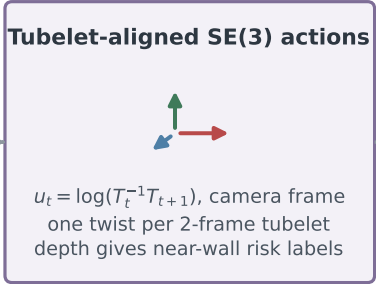
Lane B: physical grounding

audited on pose-gated data only



SCARED (laparoscopy)

frame + calibrated pose



SCARED (laparoscopy)

SE(3)-conditioned retrieval

same frozen tokens

twist u

$\hat{z}$ (risk off)

retrieval proxy

Correctness audits applied to both lanes

Input-sensitivity tests
history / action / domain

C3VD depth-warp diagnostic
convention not validated

Action negatives
batch shuffle / fixed bank

Video/patient-grouped splits
logged audit-set contacts